

TECHNICAL APPENDIX

One architecture. Three views.

Gate 5 Reference connects a working football domain, evidence exchange between independent owners, and externally authored software. This volume contains the same F, D and A sections as the standalone dossiers, preceded by the shared basis and verification guide.

Engineering basis: 64f2b38b0d671e4907bae8649fff8f3d2ee0751c

Reading routes

Route	Question
F / Football Domain	What changed between the original request and the recorded booking?
D / External DRS	How did another owner's material become a verified input?
A / Independent Authoring	What did the author receive, construct and control?
X / Shared basis	How do claims, history, publication and verification fit together?

The saved experiment uses a synthetic venue and an explicitly amended authoring protocol. G55L is published. This documentation package is prepared for independent review and subsequent G56 publication. Its own future commit is not invented.

PDF bookmarks and stable section IDs preserve the same navigation across all four documents. Source links refer to the accepted engineering commit, original archive members, or the linked publication corpus. Repository access is required while GitHub remains private.

Technical Appendix

X01 / GATE 5 REFERENCE

One accepted engineering basis

Gate 5 Reference joins three results. A football booking program carries out bounded human instructions. External DRS lets that program use material produced under another owner. An external author constructed the program from a bounded guide and a separate domain brief. These are three views of the same architecture, with separate evidence for construction, execution and admission.

G54D1 passed the finite functional examination after an explicitly authorized fourth correction in a restored continuation. The earlier G54D trial, limited to an initial submission and three corrections, remains INCOMPLETE. The amended result preserves that history. It does not retrospectively make the original budget sufficient.

G55L admitted and published the engineering result as commit 64f2b38b0d671e4907bae8649fff8f3d2ee0751c. Its sole parent is 5cef2c52fd805daedec958109caf9d1806899eb4. The admitted tree is cacb14a9574c350fb33239680abe0c3a5ede7b5e. Exactly 212 paths were added. All 4,367 preceding tracked files and their recorded permissions were preserved. The unchanged R1 route, ordinary push and independent remote readback passed.

The G55 preparation checkpoint retains its original pending language. Its acceptance-state rule makes the later exact review and landing receipts authoritative for that transition. This publication supplies the status bridge without altering those historical bytes. The new G56 documentation package is prepared for independent review. R20 and final R21 become complete only after its separately recorded publication and closure procedure.

Three distinct events

Event	Current evidence-backed state
Functional acceptance	G54D1 accepted under the explicit fourth-correction amendment
Engineering admission	G55L committed, published and independently read back
Documentation publication	This G56 package is prepared for review; publication is pending

Sources: g55l: FINAL_RESULT.json; g55l: preservation_and_landing_verification.json; Full exact source map.

X02 / GATE 5 REFERENCE

Reading the evidence at the right level

Observed or verified statements describe a particular run or a particular inspection of retained bytes. Architectural interpretations explain what those results reveal about organization and reuse. Deferred or illustrative statements describe a development path or a hypothetical deployment. The claims register keeps these classes explicit so a future possibility cannot silently become an experimental result.

The D1 native examination recorded 16 scenarios, 56 events and seven isolated mock executor starts. The separate live series obtained three real Gemini proposals and later consumed them in five native events with one additional mock start. Those two parts total 61 events and eight isolated mock starts. An effect in the main story creates three reservations. These counts do not describe physical bookings or eight repetitions of one booking.

The two D1 supplied checks each produced the same 32,702 bytes. Their SHA-256 is 0eba67d7652765c387988d760cd5daeeacecbdf33b9dc577dbe6a1cad40ed4b0. G55L later produced a different generation of canonical report: 37,389 bytes with SHA-256 d2a25ed3b1ab123912e59d096808351dbcf18f59a16229b9abcaff9f647eacff. Each report is compared with its own expected basis, not with a report of another generation.

Pure verification reads saved source and evidence. It does not recreate the candidate execution or renew old permission. A signature binds bytes to the named test key. It does not establish a real stadium identity or guarantee the truth of every assertion. The reviewed-reference trust boundary includes the exact runner, observer, examiner and frozen inputs. Arbitrary hostile Python isolation and production security certification remain outside this reference.

What each method can establish

Method	Meaning
Native observation	Actual local Work, Root decisions, dispatch, mock registry and readback in the named run
Live provider capture	Actual returned semantic fields and their later consumption
Pure supplied check	Consistency and binding of retained evidence without new runtime authority
Source and remote review	Exact admitted files, Git identities and published bytes

Sources: [g55l: canonical_result.json](#); [acceptance_matrix.json](#); [Full exact source map](#).

X03 / GATE 5 REFERENCE

Inspect saved work without performing it again

Start with the main presentation for the connected story. Choose F for the actual football process, D for external evidence and lifecycle, or A for independent construction. Each dossier retains stable section IDs in this combined appendix. GATE5_REFERENCE_READER_V01.xml is the machine entry. Its topic routes point to a shared source index and the accepted portable corpus, with explicit inclusion and omission records.

The repository contains a bounded offline route: demo/verify_gate5_reference_v01.py. Read that implementation before running it. Use Python 3.12 or later with sys.monitoring and the project dependencies already installed. The recorded owner environment used Python 3.14. Choose a fresh absolute output directory outside the repository. The command checks exact dependencies, reconstructs inert historical materials and invokes two explicitly selected pure verifier subprocesses.

No candidate import, author call, provider call, native peer, new Root decision, Work or mock action belongs to that verification route. Reading files, hashing and launching the two verification processes are real operating-system activities. The phrase offline verification does not mean zero process activity. G55L already ran the installed route and returned the exact expected canonical bytes.

The complete linked source set is larger than a single short prompt. Reader inventories state which files are inline, exact linked sources, derived explanations or intentionally omitted duplicates. Access to private repository sources requires repository permission. New publication assets are local in this bundle and become GitHub targets only after G56 publication. A later broad clean-clone exercise and a separate model comprehension examination belong to Gate 6.

Installed repository command; use a new external output path

```
PYTHONDONTWRITEBYTECODE=1 python3 -B \  
demo/verify_gate5_reference_v01.py \  
--output /absolute/new/external/proof
```

Sources: [verify_gate5_reference_v01.py](#); [g55l: canonical_result.json](#); [Full exact source map](#).

X04 / GATE 5 REFERENCE

Claims, sources and editorial method

content_manifest.json is the common publication index. It records slide, section and claim IDs and the shared prose used to build the three dossiers and this appendix. claims_evidence.json links assertions to their producing basis, checking basis, verification method and limits. evidence/SOURCE_MAP.json records exact repository or archive member identities. Repeated references to one summary do not count as independent experiments.

The retained corpus contains historical source versions as well as current admitted code. G52 calibration proofs keep their original contracts. The later G53 body-profile seam and the football program have their own source bindings. A historical source is not silently replaced with a newer implementation. Likewise, the archived B1 failure remains visible beside the revised contract and the accepted D1 continuation.

This publication uses the USENIX OSDI 2026 artifact guidance for a clear entry and separate getting-started and detailed instructions. Assertion-Evidence informs the use of a supported message and a visible example. The CMU SEI assurance-case approach informs the relationship between a claim, its argument and bounded evidence. Diataxis informs the separation of explanation and reference. These are editorial methods, not external validations of Radiolaria.

No conference badge, certification or endorsement follows from using those methods. This is a finite reference result and a reusable construction contract. Global discovery, production identity infrastructure, richer cross-domain transfer, Reflective Improvement Radiolaria and an integrated Capability Synthesis Factory remain future work. The observed path already shows how a useful result can cross an owner boundary while local decisions remain local.

Methodological primary sources, accessed 25 September 2026

Source	Use in this publication
USENIX OSDI 2026	Single entry, concrete artifact scope, getting started and detailed instructions
Assertion-Evidence	Supported headline with visual evidence
CMU SEI: Arguing Security	Claim, argument, evidence and bounded assumptions
Diataxis explanation and reference	Connected explanation separated from precise reference

Sources: [gate5_reference_contract_v01.md](#); [Full exact source map](#).

F01 / GATE 5 REFERENCE

Three training sessions became a recorded, reviewable result

Human intent → External offer → Owner consent → Recorded booking

MECHANISM OVERVIEW / Recorded evidence is cited in the sections below.

The motivation was Andrey's real trip to arrange a training camp for friends. The executed venue, inventory and prices are synthetic. The Football Domain is one of three connected Gate 5 results: a useful application, external evidence shared across owners, and an independently authored extension. The accepted program arranged three 90-minute sessions on a full-size natural-grass field after the owner permitted one narrowly defined change. It then recorded an authorized mock booking and could recover that result without repeating the effect. This dossier follows the application and its evidence rather than the chronology of building the authoring kit.

The setting is deliberately inspectable. The venue, inventory and monetary amounts are synthetic. The resulting state change is real within a disposable mock registry; it is not a reservation at an operating stadium or a payment. Two business participants, VenueRoot and ActivePlanetRoot, retain separate local decisions. The venue supplies an offer, while the requester checks whether that evidence can serve its own task.

The engineering source is admitted at commit 64f2b38b0d671e4907bae8649fff8f3d2ee0751c. Its final D1 native matrix contains sixteen accepted scenarios, and a separate live series consumes three genuine Gemini proposals. Acceptance follows an explicitly authorized fourth author correction in a restored continuation. The earlier G54D trial remains INCOMPLETE. Those qualifications identify the result accurately without changing what the working program achieved. Publication and the future Gate 6 program have their own status.

Sources: [S-EDITORIAL-MOTIVATION](#); [F-S01](#); [F-S02](#); [F-S06](#); [Full exact source map](#).

F02 / GATE 5 REFERENCE

Hard requirements make the initial failure intelligible

The request names three civil-time intervals in Europe/Tirane: 7 January 2027, 16:30-18:00; 8 January, 11:00-12:30; and 9 January, 11:00-12:30. Each session must last 90 minutes and use a full-size natural-grass field. The budget is 7,000 EUR minor units: EUR 70.00. Keeping the minor-unit representation visible prevents a hundredfold error when a technical result becomes a human story.

The main inventory includes an artificial field priced at one minor unit and a natural-grass field, field:elm, priced at 2,100 minor units per session. The cheap field cannot satisfy the surface requirement. The natural field has an occupied interval on 8 January from 11:00 to 11:30 local time, intersecting the requested session. Broad opening hours therefore do not imply that every requested interval is free.

The domain treats intervals as half-open: $\text{overlap}([a,b],[c,d))$ is true exactly when $a < d$ and $c \leq b$. A session beginning at the occupied interval's end does not overlap it. Availability, duration, surface, field size and total cost are checked together. These constraints explain why the exact request has no complete solution and why a 30-minute shift can later make one possible. The evidence exposes inventory and selection inputs, allowing the reasoning to be checked independently of the candidate's status string.

Synthetic main inventory

Field	Surface	Full size	Price / 90 min	Relevant occupancy
field:cheap	Artificial	Yes	1 minor EUR	None
field:elm	Natural grass	Yes	2,100 minor EUR	8 Jan 11:00-11:30

Sources: F-S04; F-S05; F-S22; F-S28; [Full exact source map](#).

F03 / GATE 5 REFERENCE

EXACT preserved the request and found no complete offer

The first live provider request asks for the three sessions at exact times and supplies no approval reference. Gemini returns FIND_OFFER with request revision 1 and shift allowances [0, 0, 0]. The dates, timezone, session identifiers, team, venue and 7,000-unit budget remain present in the returned parameters. The model's contribution is a concrete proposed operation and its bounded fields, rather than a claim that a pitch has already been found.

In the subsequent native live capture, both publisher and requester intake reconstruct the same normalized request. Their recorded request hash agrees with the independently preserved proposal. Each side then executes one pure Work in this FIND event. The venue's calculation has no eligible complete set, and the requester consumes the evidence for that conclusion. The independently reviewed result is NO_COMPLETE_OFFER, with no dispatch, executor start or registry mutation.

This is a useful outcome. Substituting artificial grass, shortening a session or moving it without consent would have satisfied a weaker problem than the one asked. The program retains the original requirements and exposes the missing fit. The native main matrix exercises the same exact request using controlled inputs; the live series separately demonstrates that an actual model response reached the program. These two sources support related claims but are not the same run.

Actual model user input

Find three full-size natural-grass 90-minute football sessions: 7 January 2027 16:30-18:00, 8 January 11:00-12:30, and 9 January 11:00-12:30. Exact times only. Budget 7000 EUR minor units. Request request:exam:one revision 1; team:one; venue:synthetic; session IDs session:7, session:8, session:9. No approval reference.

Actual returned text

```
{
  "operation": "FIND_OFFER",
  "parameters": {
    "request_id": "request:exam:one",
    "revision": 1,
    "team_ref": "team:one",
    "venue_ref": "venue:synthetic",
    "timezone": "Europe/Tirane",
    "currency": "EUR",
    "budget_minor": 7000,
    "owner_approval_ref": null,
    "sessions": [
      {
        "session_id": "session:7",
        "start": "2027-01-07T16:30:00",
        "end": "2027-01-07T18:00:00",
        "allow_shift_minutes": 0
      },
      {
        "session_id": "session:8",
        "start": "2027-01-08T11:00:00",
        "end": "2027-01-08T12:30:00",
        "allow_shift_minutes": 0
      },
      {
        "session_id": "session:9",
        "start": "2027-01-09T11:00:00",
        "end": "2027-01-09T12:30:00",
        "allow_shift_minutes": 0
      }
    ]
  }
}
```

Sources: [F-EXACT-REQUEST](#); [F-EXACT-TURN](#); [F-EXACT-RAW](#); [F-S02](#); [Full exact source map](#).

F04 / GATE 5 REFERENCE

One permitted change opened the offer

The second live instruction explicitly allows only the 8 January session to move by up to 30 minutes. It requests revision 2 and supplies consent:jan8-only, while stating that this is not approval to book. The returned FIND_OFFER preserves the original requested times and changes only the second session's allowance to 30. The allowance vector is [0, 30, 0], not three rewritten appointments.

That distinction separates the request from a selected solution. Native selection chooses 8 January 11:30-13:00, immediately after the occupied interval. The sessions on 7 and 9 January remain unchanged. All three use field:elm, satisfying the natural-grass and full-size requirements. At 2,100 minor EUR each, the total is 6,300 minor EUR, or EUR 63.00, within the unchanged EUR 70.00 budget. These are test amounts rather than quoted market prices.

Both parties' actual live intake hashes agree on the revised request. The source Work produces OFFER_READY, and the requesting Work consumes its checked evidence without creating a booking. The offer carries the selected times, whereas the request continues to record the original times and permission to search nearby. This preserves an auditable explanation of what changed: the human relaxed one scheduling constraint; the program selected a newly valid set within that relaxation.

Requested and selected local times - Europe/Tirane

Date	Original request	Allowance	Selected	Price
7 Jan 2027	16:30-18:00	0 min	16:30-18:00	2,100
8 Jan 2027	11:00-12:30	30 min	11:30-13:00	2,100
9 Jan 2027	11:00-12:30	0 min	11:00-12:30	2,100
Total				6,300 minor EUR

Actual model user input

For the same request and same hard requirements, I explicitly allow only the January 8 session to shift up to 30 minutes. Keep original requested times; mark its shift allowance 30, the others zero. Use revision 2 and local approval reference consent:jan8-only. This does not approve a booking.

Previous normalized request: {"budget_minor": 7000, "currency": "EUR", "operation": "FIND_OFFER", "owner_approval_ref": null, "request_id": "request:exam:one", "revision": 1, "sessions": [{"allow_shift_minutes": 0, "end": "2027-01-07T18:00:00", "session_id": "session:7", "start": "2027-01-07T16:30:00"}, {"allow_shift_minutes": 0, "end": "2027-01-08T12:30:00", "session_id": "session:8", "start": "2027-01-08T11:00:00"}, {"allow_shift_minutes": 0, "end": "2027-01-09T12:30:00", "session_id": "session:9", "start": "2027-01-09T11:00:00"}], "team_ref": "team:one", "timezone": "Europe/Tirane", "venue_ref": "venue:synthetic"}

Actual returned text

```
{"operation":"FIND_OFFER","parameters":{"request_id":"request:exam:one","revision":2,"team_ref":"team:one","venue_ref":"venue:synthetic","timezone":"Europe/Tirane","currency":"EUR","budget_minor":7000,"owner_approval_ref":"consent:jan8-only","sessions":[{"session_id":"session:7","start":"2027-01-07T16:30:00","end":"2027-01-07T18:00:00","allow_shift_minutes":0},{session_id":"session:8","start":"2027-01-08T11:00:00","end":"2027-01-08T12:30:00","allow_shift_minutes":30},{session_id":"session:9","start":"2027-01-09T11:00:00","end":"2027-01-09T12:30:00","allow_shift_minutes":0}]}}
```

Sources: [F-S04](#); [F-S08](#); [F-S02](#); [F-S30](#); [Full exact source map](#).

F05 / GATE 5 REFERENCE

The offer became an input to another owner's Work

The venue's result does not become local authority merely by crossing a boundary. In the accepted FIND path, VenueRoot owns the producer Work and its review. The published offer is disclosed through the governed external exchange. ActivePlanetRoot checks the received body, source status, task binding and local policy before its own Work uses that evidence. The two Roots are peers; the runner and examiner are instruments of the experiment, not a third business owner above them.

Three representations describe one causal history. E is the producer Work's domain result. B is the published body containing E together with source, Work, review and time references. C is the checked consumption context containing B and the later status information needed by the recipient. The required relationships are $B.offer = E$ and $C.body = B$, not byte equality among all three documents. Added evidence explains a transformation rather than silently replacing the original result.

The main SHIFT capture contains the producer Work, offer body, consumption context and requester Work as distinct exact files. Independent review joins them to actual recorded events and confirms local consumer execution. External material therefore became an input to work owned by the other side. The External DRS dossier explains the address, retrieval and lifecycle mechanisms in detail; this example shows their application to a football offer.

Sources: [F-S06](#); [F-S07](#); [F-S08](#); [F-S09](#); [Full exact source map](#).

F06 / GATE 5 REFERENCE

Separate consent led to one three-slot state change

Finding a valid offer is not approval to execute it. BOOK is a separate controlled owner input with operation CONFIRM MOCK BOOKING and reference consent:controlled:booking. Local policy binds that approval to the booking request and preceding offer request. The semantic provider does not manufacture this permission, and the January 8 shift reference cannot substitute for it.

The venue's reviewed native route checks current policy, source state, offer and request binding, schedule revision, tariff, occupancy and the existing registry before the effect. The main event records one Host dispatch, one executor start, one effect callback and one business-state write. That single mock action adds three reservation records and one idempotency entry in the owning registry. It is one batch effect with three slots, not three separately authorized executions.

Evidence goes beyond the returned MOCK_BOOKED label. The saved native receipt is an EVIDENCE_ONLY artifact owned by VenueRoot; it records the effect result rather than granting permission for a future one. The registry snapshot and independently observed final source state agree. The receipt, action result and readback are joined by the examiner to the actual trace. This establishes an observed state change in the synthetic registry. It does not establish a transaction with a real venue or resilience to untested operating-system failures.

Five events in the native main history

Event	Operation	Observed result	New mock effects
EXACT	FIND_OFFER	NO_COMPLETE_OFFER	0
SHIFT	FIND_OFFER	OFFER_READY	0
BOOK	CONFIRM MOCK BOOKING	MOCK_BOOKED	1
REPEAT	CONFIRM MOCK BOOKING	Same receipt	0
VERIFY	VERIFY_EXISTING	VERIFIED	0

Sources: F-S05; F-S06; F-S07; F-S12; [Full exact source map](#).

F07 / GATE 5 REFERENCE

Repeat and verification recovered the result without executing it again

The main REPEAT event repeats the confirmed booking request after the first effect has completed. It returns the previous receipt instead of creating another reservation batch. Independent flow review records zero new pure Work, dispatches, executor starts, effects and state writes for both roles. The registry hash remains equal to the post-BOOK snapshot, so an unchanged report is supported by unchanged business state.

VERIFY has a different operation: VERIFY_EXISTING. Its actual live model input asks to use existing history and readback without searching or booking again. Gemini returns that operation with revision 2, the same session constraints and the historical booking approval reference. Later native intake consumes it on both sides. The result is VERIFIED, with the existing receipt and no new search Work or effect. The reference is historical evidence, not a fresh permission to act.

The timing matters. All three live proposals, including VERIFY, were collected before the subsequent native booking run. The model did not see the receipt that the program later created. What was demonstrated is preparation and later consumption of a request to verify, followed by the program's actual historical check. The saved evidence supports this precise sequence without inventing an interactive conversation in which the model personally observes the completed booking.

Actual model user input

Verify the already completed booking from source history and readback only. Do not search for or book anything new. Keep revision 2, the same sessions and January 8 allowance. Historical booking approval reference is consent:controlled:booking, not a new grant.

```
Previous normalized request: {"budget_minor": 7000, "currency": "EUR", "operation": "FINDER", "owner_approval_ref": "consent:jan8-only", "request_id": "request:exam:one", "revision": 2, "sessions": [{"allow_shift_minutes": 0, "end": "2027-01-07T18:00:00", "session_id": "session:7", "start": "2027-01-07T16:30:00"}, {"allow_shift_minutes": 30, "end": "2027-01-08T12:30:00", "session_id": "session:8", "start": "2027-01-08T11:00:00"}, {"allow_shift_minutes": 0, "end": "2027-01-09T12:30:00", "session_id": "session:9", "start": "2027-01-09T11:00:00"}], "team_ref": "team:one", "timezone": "Europe/Tirane", "venue_ref": "venue:synthetic"}
```

Actual returned text

```
{"operation": "VERIFY_EXISTING", "parameters": {"request_id": "request:exam:one", "revision": 2, "team_ref": "team:one", "venue_ref": "venue:synthetic", "timezone": "Europe/Tirane", "currency": "EUR", "budget_minor": 7000, "owner_approval_ref": "consent:controlled:booking", "sessions": [{"session_id": "session:7", "start": "2027-01-07T16:30:00", "end": "2027-01-07T18:00:00", "allow_shift_minutes": 0}, {"session_id": "session:8", "start": "2027-01-08T11:00:00", "end": "2027-01-08T12:30:00", "allow_shift_minutes": 30}, {"session_id": "session:9", "start": "2027-01-09T11:00:00", "end": "2027-01-09T12:30:00", "allow_shift_minutes": 0}]}
```

Sources: [F-S06](#); [F-S13](#); [F-S14](#); [F-S15](#); [Full exact source map](#).

F08 / GATE 5 REFERENCE

Three genuine model responses have a traceable consumer

The live provider is gemini-2.5-flash. The attempt ledger records exactly three completed requests, one wire send per role, for EXACT, SHIFT and VERIFY. The inputs disclose the method: a supplied system instruction limits the response to an operation and parameters; a JSON schema enumerates request fields; SHIFT and VERIFY also receive the previous normalized request. Temperature is 0.0, maximum output is 4,096 tokens, and returned content is JSON. Field choice, price, booking success and Root authority are outside the model's proposed decision.

Consumption is checked at both business roles. EXACT intake returns occur at observer sequence 5 for publisher and requester. SHIFT returns occur at 31 and 61; VERIFY at 106 and 189. In each pair, the reconstructed intake request hash equals the corresponding normalized provider request hash. The verifier also checks parent events and captured intake values before joining to independent Work or history results. An HTTP success response alone would not establish this chain.

The contribution is specific and useful: a changed human instruction changes the operation or a permitted parameter, and the actual program follows it. Native Work determines the offer; local policy and Root determine admissibility; controlled owner input supplies booking consent. The complete input, returned text and schema are retained alongside this dossier as exact source material. No new model call was made to prepare the publication.

Actual proposal, native intake and result

Role	Operation / allowance	Publisher / requester intake	Native result
EXACT	FIND_OFFER / [0, 0, 0]	5 / 5	NO_COMPLETE_OFFER
SHIFT	FIND_OFFER / [0, 30, 0]	31 / 61	OFFER_READY
VERIFY	VERIFY_EXISTING	106 / 189	VERIFIED

Actual shared system instruction

Interpret the user's intent into a bounded football semantic proposal. Return only JSON operation and parameters. No field selection, prices, booking success or Root authorization is yours to decide. Preserve identifiers and civil times exactly. Datetime strings use YYYY-MM-DDTHH:MM:SS, Europe/Tirane. A shift allowance means keep original times and change only allow_shift_minutes for January 8. Parameters include all request fields except operation. Operations: FIND_OFFER, VERIFY_EXISTING, CLARIFY. An owner approval reference is a supplied local reference, never permission created by you.

Sources: [F-S02](#); [F-S03](#); [F-S29](#); [F-EXACT-REQUEST](#); [Full exact source map](#).

F09 / GATE 5 REFERENCE

Sixteen cases exercise one final program

The final native matrix executes the same frozen D1 candidate across sixteen scenarios. The complete rows are reproduced here rather than replacing the matrix with a single PASS total. They account for 56 events and seven isolated mock executor starts. The main case contributes five events and one effect; other positive and variation cases have their own disposable registry and evidence. Seven effects are not seven repeats of the main booking.

The cases cover useful work, permissions, source suitability, input variation and history. A low budget may legitimately produce no complete offer. Missing BOOK approval prevents action. Missing history yields CURRENT_STATUS_UNKNOWN. Revoked or wrongly addressed external material cannot become usable local evidence. A changed schedule revision or occupancy blocks later confirmation even when the earlier FIND produced an offer.

Other cases exercise reordered inputs, changed identifiers and prices, another team and lawful continuation after earlier refusals. The same-key-changed case contains a successful booking followed by refusal of a changed repeat, explaining its one effect. The final reserved variation is the preserved G54D challenge, not a newly unseen benchmark after restored continuation. Together the cases establish breadth within this finite domain and fixture family; they do not certify a universal planner for every training-camp problem.

Complete recorded D1 native matrix

Case	Events	Bookings	Host dispatches	Executor starts	Result
main	5	1	1	1	PASS
low_budget	1	0	0	0	PASS
missing_BOOK	3	0	0	0	PASS
missing_history	1	0	0	0	PASS
release_denied	2	0	0	0	PASS
revoked_source	2	0	0	0	PASS
foreign_audience	2	0	0	0	PASS
revision_conflict	3	0	0	0	PASS
same_key_changed	4	1	1	1	PASS
permuted_main	5	1	1	1	PASS
variation_feedback	5	1	1	1	PASS
second_team	5	1	1	1	PASS
long_overlap	2	0	0	0	PASS
occupancy	3	0	0	0	PASS
continuation	8	1	1	1	PASS
variation_final_reserved	5	1	1	1	PASS

Sources: [F-S01](#); [F-S06](#); [F-S17](#); [F-S19](#); [Full exact source map](#).

F10 / GATE 5 REFERENCE

Refusal is part of a complete domain process

The most consequential final correction concerns a source whose authenticated status is REVOKED. The program already stopped using that source without booking, but the earlier G54D candidate omitted the required local Root refusal in this exception branch. D1 retains the stop and now records a genuine BLOCKED_FAIL_CLOSED decision, bound to the original request, source record, revision, status and reason status_not_active. The evidence identifies actual decision calls, not a newly written refusal caption.

For the two revoked FIND events, the recorded refusal calls occur at sequences 29 and 66. Both are joined to the authenticated terminal source state and corresponding request hashes. The final native case has two events, no dispatch and no mock executor start. The old trial remains INCOMPLETE: an explicitly authorized extra correction repaired a real profile obligation without rewriting the earlier result.

Other negative cases address distinct causes rather than collapsing them into one generic failure. Missing history is uncertainty, denied disclosure is a release refusal, foreign audience is a scope mismatch, and an occupied or changed schedule prevents later action. Correct behavior is assessed using the concrete inputs and captured path. A program that could only display a successful booking would miss these boundaries; this domain also makes the reasons for withholding action inspectable.

Sources: [F-S01](#); [F-S16](#); [F-S17](#); [F-S24](#); [Full exact source map](#).

F11 / GATE 5 REFERENCE

Variation changes the answer without replacing the program

The reference keeps the authored source fixed while changing facts around it. Permuted inputs test independence from presentation order. Changed inventory and pricing test whether selection depends on actual source material. Another team tests request identity and task binding. The continuation case includes earlier refused events followed by a later admissible request and one successful booking, preserving the distinction between a scoped refusal and a permanent ban on the whole task.

Recorded variant results make that dependence visible. The final reserved variation totals 6,798 minor EUR, rather than the main case's 6,300. The second-team and continuation accepted offers total 6,414. These are results of their own supplied inventories and requests, not alternative prices retrospectively attached to the main story. Each successful path has its own receipt and source-state readback, and repeats are checked within that path's history.

Nine coherent altered-evidence controls form a separate pure examiner test group. They ask whether internally edited evidence is rejected; they are not nine additional native attacks. Native refusal scenarios likewise do not demonstrate every possible race or failure between validation and execution. The useful architectural conclusion is bounded: a new authored domain can use the same governed activity and exchange mechanisms across this range of changed inputs, with acceptance grounded in records of what actually happened.

Sources: [F-S01](#); [F-S18](#); [F-S19](#); [Full exact source map](#).

F12 / GATE 5 REFERENCE

The saved result can be checked without booking again

Three evidence levels remain separate. D1 native acceptance covers sixteen controlled scenarios, 56 events and seven isolated mock effects. Its live series adds three genuine provider responses, a separate five-event run and one additional mock effect. Across these groups there are 61 events and eight mock effects, not eight payments or one booking repeated eight times. The authoring model, gpt-6-astra, created the program during a different process; it is not the runtime semantic provider.

The independent D1 review verified 362 Ed25519-signed messages in the native corpus, 25 completed FIND joins across E, B and C, and seven confirmation consumptions. These are scoped native evidence checks, not counts of Roots or every cryptographic operation across Gate 5.

The two accepted D1 supplied results are each 32,702 bytes and have SHA-256 0eba67d7652765c387988d760cd5daeeacecbdf33b9dc577dbe6a1cad40ed4b0. They recheck saved source-bound evidence in separate pure processes. The installed Gate 5 verification entry reconstructs declared historical inputs and invokes reviewed pure consumers, without importing the candidate, calling a provider, making new Root decisions or performing a mock effect. File reads, hashing and verification processes still occur.

Trust is limited to the exact reviewed reference implementation, observer, runner, examiner and pinned evidence. This does not attest arbitrary hostile Python, real venue identity, production payments, all prestart races, post-effect uncertainty or power-loss recovery. Keeping receipts evidence-only prevents history from becoming current permission. Within that scope, the dossier traces original human conditions through model parameters, actual Work, authorized mutation and independently checked history. The adjoining DRS and authoring dossiers explain the other two views of the same engineering result.

Sources: [S-REVIEW-G54D1-INDEPENDENT](#); [F-S01](#); [F-S02](#); [F-S03](#); [Full exact source map](#).

D01 / GATE 5 REFERENCE

External evidence became useful without changing who could decide

Owner-held material → Address + metadata → Permitted payload → Local Work

ARCHITECTURAL MECHANISM / Addresses and material remain distinct.

External DRS is the Gate 5 view of how one owner's work becomes usable material for another owner. In the controlled calibration case, a source owner computes a correction; the receiving owner obtains an allowed fragment, checks its origin and current applicability, and consumes it in a different calculation. The football domain applies the accepted exchange with a different content profile. The result is a measured relationship between separate work, rather than two reports placed beside each other.

This dossier accompanies the engineering basis 64f2b38b0d671e4907bae8649fff8f3d2ee0751c. Its concrete evidence comes from the G52 lifecycle, the G54C numerical contract probe and the accepted G54D1 football program. Their roles differ: G52 demonstrates a two-process exchange and restart; G54C isolates representation and consumption; D1 demonstrates the authored domain. Historical checkpoints keep their original preparation status. They are interpreted through the later engineering admission, not silently rewritten.

The architectural point is that evidence and authority travel differently. A signed result can influence a receiving computation while its publisher retains control of disclosure and its recipient retains control of local use. Neither side becomes the other's superior. Gate 5 establishes this finite reference with controlled data and pinned test identities. It does not claim a deployed public discovery network or production venue integration.

Sources: [gate5_reference_contract_v01.md](#); [gate5_reference_checkpoint_v01.md](#); [gate5_lifecycle_restart_checkpoint_v01.md](#); [g54c: summary.json](#); [Full exact source map](#).

D02 / GATE 5 REFERENCE

The addressable structure is different from the material it describes

Local archive → Descriptor / index → Allowed payload → Verified context

CONCEPTUAL VIEW / The index may carry permitted metadata and summaries.

DRS names a semantic topology: relationships between experience, provenance, time, trust, conflicts and possible reuse. A rich local record can include the result of work, its supporting traces and the context in which it was valid. The addressable description helps another task discover which material might matter. It does not require every index to contain the owner's complete archive, and it does not make an indexed record a local decision.

Four distinct places make the mechanism legible. First, the source owner holds the detailed material. Second, a descriptor and permitted metadata identify a bounded object in an address index. Third, an approved retrieval delivers a payload. Fourth, the receiver constructs a checked context for a particular Work. Metadata can contain an allowed summary; "pointer only" means the payload has not already been smuggled into the pre-retrieval index, not that the descriptor contains no information at all.

The historical architecture documents use a DNS-like analogy for semantic addressing. This is an explanatory analogy, not DNS wire compatibility. The G52 capture provides the narrower observable example: a persisted LocalDRS lookup exists before the body is received. The receiver may subsequently preserve an accepted fragment and its provenance locally. That useful retention is compatible with ownership remaining separate. It is not evidence that all participants share one global memory database.

Sources: [passport_geometry_root_needles.md](#); [human_passport_v0_25.md](#); [strategic_expansion_map.md](#); [g52: before_body.json](#); [Full exact source map](#).

D03 / GATE 5 REFERENCE

Finding an address, receiving a body and using it have different grounds

A useful address starts a question: may this local task retrieve and use the referenced material? In the reference exchange, the requester’s local policy and Root review govern its own retrieval. The source separately authenticates the request and evaluates disclosure for the declared requester, object, purpose and use. Knowing a pointer therefore does not entitle the caller to its body. The source’s release decision does not decide the receiving owner’s subsequent computation.

The actual G52 peers have separate processes, local state, keys and Root reviews. They exchange signed messages through inherited OS pipes. The public trust configuration pins the expected identities; the received object cannot substitute its own trust anchor. Source, publication and release reviews have different subjects. On the receiver, retrieval, import and result review likewise bind different stages. The supervisor and examiner observe the experiment; they are not a third business owner above the two Roots.

The refusal contrasts make the division observable. A valid pointer can encounter a local refusal without a body fetch. An authenticated caller can also be refused by the publisher before body disclosure. These are different boundaries, with different counter evidence. Request scope and budget remain attached to the declared context. The successful neighbor then shows that a permitted fragment can enter useful work without turning either signature or remote acceptance into local permission.

Who controls the next step

Stage	Local question
Requester retrieval	May this task request this material?
Publisher release	May these bytes be disclosed to this caller and use?
Requester consumption	Is this evidence applicable to this current work?

Sources: [gate5_lifecycle_restart_checkpoint_v01.md](#); [g52: before_body.json](#); [g52: result.json](#); [gate5_lifecycle_supplied_v01.py](#); [Full exact source map](#).

D04 / GATE 5 REFERENCE

Evidence gained context without changing its original result

E: Work result → B: E + provenance → C: B + status → Receiving Work

CAUSAL REPRESENTATIONS / B.offer = E; C.body = B.

The shared contract separates three causal objects. E is the domain result extracted from a real producer Work. B is the later published representation: it contains the result together with source identity, revision, time and links to the completed Work and its review. C is the receiver's consumption context, constructed after checking the exact publication, release and current signed status against local policy and the current request. Later checks can add information that did not exist when E was computed.

This ordering matters. A producer cannot return the identifier of its own future review as an already established fact. Nor can an earlier published body contain a later status observation. The correct condition is preservation of meaning across representations: `payload_projection(B)` equals E, and C retains B while adding independently derived context. In the football profile, B.offer is the produced offer and C.body is the published B. The objects have different roles and need not be byte-identical.

The examiner reconstructs expected C from original source inputs, the actual producer result and review, transmitted body, authenticated status, and independently held local observations. It then checks what the consumer truly received and returned. Copying the candidate's own declared input into the expected input would only prove self-consistency. Gate 5 instead requires the causal joins. This general contract was corrected visibly after the earlier B1 mismatch; the old incomplete result remains part of the history.

Sources: [SHARED_EVIDENCE_CONTRACT.md](#); [g54d1: review_finds.py](#); [gate5_reference_contract_v01.md](#); [Full exact source map](#).

D05 / GATE 5 REFERENCE

A small numerical probe tested the common contract outside football

G54C exercised the E/B/C relationship with a deliberately small numerical example. The producer evaluates $3 \times x + 2$ for inputs 7 and 11, yielding 23 and 35. The numerical body carries this computed value and the original sample. After public component checks, the consumption context adds the checked request revision and source/status projection. The consumer computes $B.computed + B.sample + request_revision$. With revision 1, the actual results are 31 and 47.

The arithmetic is easy to inspect, but the causal requirement is stronger than arithmetic agreement. The published computed field must come from the recorded producer Work, its source Work and review references must match real records, and the status must belong to this pointer and request. A correct consumer calculation over a coherently fabricated context is still the wrong computation for the independently established input. The separate examiner checks that distinction.

The preserved probe includes nine negative controls: another signed payload, Work reference or review reference; stale or differently bound status; a missing consumed field; and altered context or result projection. They fail their corresponding predicates while the unchanged positive remains valid. These are recorded component and evidence controls. The probe used actual producer/review/consumer Work, but peer_io and effects are zero: it is not another two-peer exchange or an action authorization. Its value is isolating a reusable representation contract before applying it to the larger domain.

Recorded numerical probe

Input x	Producer $3x+2$	Consumer $E+x+1$
7	23	31
11	35	47

Sources: [SHARED_EVIDENCE_CONTRACT.md](#); [numeric_context.py](#); [g54c: summary.json](#); [Full exact source map](#).

D06 / GATE 5 REFERENCE

New external evidence changed 96 into 102 while the local rule stayed fixed

The separate G52 calibration story demonstrates actual external consumption. Owner A has a reference value and observations; Owner B has its own readings and assessment limit. A’s correction is reference minus the mean of its observations. At source revision 1, reference 100 and observations [110, 110, 110] produce -10. B’s readings [103, 106, 109] have mean 106. Applying the received correction in typed local Work produces 96, within B’s unchanged limit of 100.

At source revision 2, A’s observations become [104, 104, 104]. Its new native Work produces -4. B retains the same readings and local limit, obtains the new permitted source, and computes 102. The accurate assessment is now REVIEW_REQUIRED. A local Root can accept that faithful report without declaring the measurement within limits. Acceptance of a result and a favorable business assessment are distinct statements.

The saved 96 is not edited into 102. The second result has its own immutable source, lineage, adaptation and current review, while the first stays historical. The uncalibrated mean does not receive a canned correction; missing calibration is refused, and a separate input contrast produces 97. These are controlled numerical observations, not a physical sensor deployment. They show exactly what changed: the imported computed basis affected the receiving calculation, while the receiving rule and its earlier evidence remained intact.

The local inputs and limit are unchanged

Source revision	A observations	A correction	B result / limit
1	110,110,110	-10	96 / 100
2	104,104,104	-4	102 / 100

Sources: g52: [accepted.json](#); g52: [accepted.json](#); [gate5_lifecycle_restart_checkpoint_v01.md](#); g53: [result.json](#); [Full exact source map](#).

D07 / GATE 5 REFERENCE

Freshness belongs to the source, the status and the current use

The accepted external profile has an explicit interval for source usability. Its end is the earliest of `valid_to`, creation time plus TTL, and source observation time plus the receiver's maximum allowed source age. A use must fall inside the half-open interval: `valid_from ≤ use_time < end`, with the source observation no later than `use_time`. Equality at the end is expired. Local ingestion cannot refresh the age of an old observation.

Status freshness is checked separately. The signed observation binds the pointer, publisher and status stream to the requester, request revision and nonce. Its `checked_at` must not be in the future, and `use_time` must precede both `valid_until` and `checked_at + 60` in this finite profile. A current status cannot extend an expired source; a still-live source cannot supply a missing current status. The request and local policy introduce further applicability conditions.

The donor numerical Work uses its existing controlled logical clock; the external envelopes, current reviews and consumption use separately recorded integer UTC. This distinction avoids rewriting a kernel clock to make an example pass. In football, the civil date of a training session is another subject: a January calendar slot does not itself establish that today's offer or status is current. Supplied verification checks the recorded use time. It does not renew historical evidence for a new action.

Finite-profile validity rules

Object	Required relation
Source	<code>end = min(valid_to, PT+TTL, observed_at+max_age)</code>
Use	<code>valid_from ≤ use_time < end</code>
Status	<code>checked_at ≤ use_time < min(valid_until, checked_at+60)</code>

Sources: [gate5_contracts_v01.py](#); [SHARED_EVIDENCE_CONTRACT.md](#); [gate5_calibration_exchange_checkpoint_v01.md](#); [gate5_lifecycle_restart_checkpoint_v01.md](#); [Full exact source map](#).

D08 / GATE 5 REFERENCE

Revocation survived restart while earlier success remained readable

REVOKED saved → B process restarts → Old ACTIVE refused → New basis reviewed

RECORDED SEQUENCE / Persistent status prevents this tested rollback.

The lifecycle separates authenticating a status from deciding that it permits current use. A genuine terminal observation can be valid evidence precisely because it says the source is no longer active. In G52, the receiver authenticates and persists the REVOKED entry before refusing further current use. The high-water record belongs to that publisher, pointer and status stream; an older signed ACTIVE response cannot erase what has already been learned.

The restart is an actual process transition. B exits and is reaped while A remains alive. A new B process loads the existing pointer index, budgets, imports and history. The declared bootstrap repins the new B signer without exporting the previous private key, changing local policy or resetting task state. The previously saved ACTIVE response is then rejected as `status_rollback`. A later lawful source revision has a distinct pointer/status stream and supports a new reviewed result.

History answers a different question. Three ordinary historical queries preserve the earlier result and each perform one fresh local read review, with zero remote sends and zero new Work. The supplied verifier is more restricted still: it makes no new Root decisions. Neither path reactivates an old permission. The experiment proves the stated persisted restart behavior, not protection against rolling back an entire machine snapshot or losing durable state after power failure.

Sources: [gate5_lifecycle_restart_checkpoint_v01.md](#); [g52: restart_checkpoint.json](#); [g52: restart_old_active.json](#); [g52: terminal_observation.json](#); [Full exact source map](#).

D09 / GATE 5 REFERENCE

Budgets and negative experience limit exploration without erasing lawful continuation

The source enforces its disclosure budget before constructing a permitted body response. G52 declares operator task contexts in advance, binding requester, object, purpose, permitted use and source revision. An arbitrary new label cannot create a fresh allowance. Reopening the budget helper loads the same durable state, and the exhausted original task remains exhausted after the requester restarts. Attempt reservation and body count/byte reservation protect different points in the sequence.

The baseline G52 task records twelve attempts: eight permitted metadata operations, two permitted payload attempts and two denied payload attempts. Two bodies were disclosed, totaling 2,146 bytes. The two additional authenticated fetches were refused at the publisher before body disclosure, including after its budget helper reopened. These are G52 transport figures, not football counters. The refusal boundary is before disclosure of a body, not before every form of I/O.

Negative experience also stays scoped. An unavailable current status becomes UNKNOWN rather than assumed ACTIVE or an eternal blacklist. A known revocation blocks its current source stream; a lawful new source can support continuation under new review. Duplicate delivery leaves one source and one history record with zero credit at that point. This is bounded deduplication, not a global reputation algorithm. The broader DeadEnd concept can organize failure experience, but the measured claim here remains the specific refusal, history and continuation relationships in the capture.

G52 baseline task accounting

Counter	Recorded value
Attempts	12
Permitted metadata / payload	8 / 2
Denied payload	2
Disclosed bodies / bytes	2 / 2,146

Sources: [g52: publisher_budget.json](#); [g52: dedup.json](#); [g52: status_unavailable.json](#); [gate5_lifecycle_restart_checkpoint_v01.md](#); [Full exact source map](#).

D10 / GATE 5 REFERENCE

The negative controls reach different semantic boundaries

The F01-F20 inventory is a map of questions, not twenty interchangeable success labels. Some answers come from the actual G52 lifecycle: a pointer before a body, independent refusals, changed local conditions, terminal-state persistence, unknown status, deduplication and restart. Others come from coherent signed component mutations or bounded state controls. A correctly signed but semantically wrong object must reach the intended scope or arithmetic boundary rather than fail only because its JSON is broken.

The evidence distinguishes a tampered byte sequence from a signed wrong calculation, a foreign recipient from an unknown signer, and an expired observation from a rollback of status revision. Positive neighbors remain important: the same allowed object or lawful continuation must still succeed. Missing imported fields and the recorded result 97 provide causal checks against a receiver returning a fixed answer regardless of its actual input.

G53’s forty-three affected default controls support the later body-profile seam; they remain recorded stage evidence, not newly executed G55 tests. G55 preserves the exact historical G52 source for its supplied proof and checks the accepted D1 source separately. This prevents a newer contract being silently substituted into an old collection. The inventory is finite. Pure altered-evidence controls do not certify physical crashes, arbitrary concurrency, hostile Python execution or an internet-wide discovery service.

F01-F20 evidence families

IDs	Boundary / principal method
F01-F03, F18	Lookup, separate refusals and restart / recorded native lifecycle
F04-F06, F13	Integrity, signed scope/arithmetic, identity / component controls
F07-F12	Request, time, policy, status and conflicts / mixed native and pure
F14, F16-F17	No executable authority; bounded size, route and depth / pure controls
F15	Duplicate source/history accounting / recorded lifecycle
F19	Consumed-input dependence / missing-field control plus native97
F20	Private-note absence / recorded public-byte canary scan

Sources: [acceptance_matrix.json](#); [gate5_lifecycle_restart_checkpoint_v01.md](#); [g52: result.json](#); [g53: result.json](#); [Full exact source map](#).

D11 / GATE 5 REFERENCE

The saved work can be checked without performing it again

The installed proof command is `demo/verify_gate5_reference_v01.py` with a new output directory outside the repository. It checks pinned installed sources and portable evidence, reconstructs the recorded author's text operations, and starts two explicitly selected pure verifier subprocesses. One verifies historical G52 evidence with its exact historical contracts. The other verifies the accepted D1 native and live captures. It does not start the candidate, peers, author, provider, new Root decisions, native Work or effects.

Two G55 preparation processes independently produced the same 37,389-byte canonical result, with SHA256 `d2a25ed3b1ab123912e59d096808351dbcf18f59a16229b9abcaff9f647eacff`. Their component outputs also match the accepted recorded G52 and D1 results. The G52 consumer checks signed wires, actual Work/result structures, Root relationships, source revisions, history and restart. The D1 consumer checks its frozen cases and observed boundary records. Hash agreement is one layer; the selected semantic verifiers perform the relational checks.

The trust profile includes exact reviewed reference code, runner and observer, operator-pinned inputs and keys, and the independent examiner. Signed bytes are not encrypted transport, and test keys do not establish the identity of a real venue. The same-user IPC experiment does not claim resistance to an OS administrator. Verification reads evidence and launches pure processes; calling that zero OS activity would be misleading. This publication inspects existing evidence rather than collecting another execution.

Recheck the admitted evidence

Requirement	Value
Entry	<code>demo/verify_gate5_reference_v01.py --output /absolute/new/external/proof</code>
Runtime	Python3.12+ with sys.monitoring and project dependencies
Class	Historical G52 and accepted D1 supplied verification

Sources: [verify_gate5_reference_v01.py](#); [comparison.json](#); [gate5_lifecycle_supplied_v01.py](#); [g52: supplied_03.json](#); [Full exact source map](#).

D12 / GATE 5 REFERENCE

Independent systems could cooperate through addressable experience

Published description → Bounded discovery → Permitted retrieval → Local work

FUTURE EXTENSION / Discovery and federation are not completed in G5.

The next architectural step starts with the measured relationship, not an assertion that a global network already exists. Gate 5 showed two owners, bounded addresses, permitted payloads, currentness checks, real consumption and preserved history. A wider system could let organizations publish limited descriptions of verified experience or capabilities so that another task can find relevant material before deciding whether to request it.

Consider an illustrative future collaboration. A venue exposes a description of its available services; a laboratory describes an evaluation method; a supplier publishes a validated maintenance procedure. A task can discover the related addresses, request the necessary fragments and use them under its own local policy. No participant has to deliver its entire archive or become the common authority. The example describes an extension path, not an existing customer, integration or public catalogue measured in Gate 5.

A searchable public DRS would still need concrete discovery rules, routing, profile compatibility, identities, disclosure policies, budgets and cycle limits. Those deployment choices cannot be inferred from a working two-peer bridge. The point of the reference is to make their boundary visible: an address can introduce evidence, while each owner separately controls what is disclosed and what is done with it. The observed connection should be drawn solid; wider discovery and federation should be drawn as explicitly future connections.

Sources: [passport_geometry_root_needles.md](#); [strategic_expansion_map.md](#); [gate5_reference_contract_v01.md](#); [acceptance_matrix.json](#); [Full exact source map](#).

D13 / GATE 5 REFERENCE

New meaning is produced by work and can become new addressable experience

Existing experience → New local work → Checked result → New address

ARCHITECTURAL EXTENSION / Publication remains an owner decision.

DRS can connect existing experience to a new task, but the index itself does not generate a new conclusion. The performing Work creates the new result. Its source relationships and local review establish how that result was obtained and how it may be used. If the owner's publication policy permits, a bounded description and supporting material can then become available to another task. This closes a conceptual loop between activity, memory and later activity.

The architecture allows rich experience behind an address. A future artifact might contain a detailed trace or a fuller fractal snapshot with the relationships needed to understand an outcome. The historical passport describes boundary snapshots as addressable experience. That concept should not be confused with the small calibration payload actually transferred in G52, or with permission to execute a serialized tree. Richer representation increases what can be inspected; it does not inherit the producer's right to act.

The observed calibration revisions offer a small foothold for this direction. A new source supports a new conclusion while its predecessor remains historical, with explicit lineage instead of silent replacement. Extending that pattern across domains would require suitable content profiles and validation for each meaning. The future promise is therefore cooperative accumulation of useful, attributable experience. It is not automatic truth, universal trust or a claim that every recorded trace can be safely reused in every later context.

Sources: [human_passport_v0_25.md](#); [passport_geometry_root_needles.md](#); [strategic_expansion_map.md](#); [g52:accepted.json](#); [Full exact source map](#).

D14 / GATE 5 REFERENCE

Descriptions of capabilities can travel while admission remains a separate act

Addressable material can describe what a service is able to do: its inputs and outputs, required confirmations, allowed actions, receipts, versions and revocation conditions. A consumer could inspect such a description and use a capability at its owner’s service without downloading that service’s implementation. If a new local implementation is needed, the resulting candidate must follow a separate source review and admission path. A discovered description is not an executable import instruction.

This distinction connects the External DRS view to independent authoring. Gate 5 separately demonstrated construction of a football program from a bounded kit and its later engineering admission. It did not demonstrate a runtime discovering a service, spawning an author and installing the resulting program automatically. The accepted local body-profile seam is selected in reviewed code; a foreign payload cannot name a Python module and appoint it as its own validator.

The accepted naming map makes later directions explicit. Reflective Improvement Radiolaria concerns lessons and improvement proposals from experience. Cross-Domain Transfer Radiolaria concerns structural analogies and transfer candidates. Capability Synthesis Factory concerns an integrated requirements, code and sandbox cycle. These directions, global discovery and broader federation remain deferred after Gate 6 publication. They are future proposal mechanisms, not additional authorities. The current result is a concrete foundation: a useful result can become addressable beyond its producing system while ownership, current permission and local admission remain explicit. Cognitive Radiolaria is also a deferred umbrella direction, not a completed Gate 5 subsystem.

Separate extension paths

Path	Boundary
Use an external capability	Service remains at its owner; current local permission required
Create a local implementation	Candidate undergoes separate review and admission
Discover wider experience	Future discovery, identity and profile infrastructure

Sources: [strategic_expansion_map.md](#); [gate5_body_profile_v01.py](#); [gate5_reference_contract_v01.md](#); [acceptance_matrix.json](#); [Full exact source map](#).

A01 / GATE 5 REFERENCE

Another author built a working domain without acquiring admission authority

HOW + WHAT

→ External author

→ Independent exam

→ Owner admission

DESIGN TIME / Authoring and execution are separate processes.

Gate 5 asks whether Radiolaria's way of constructing governed work can be transferred to a developer who did not build its core. The accepted example is a football booking domain written by an external model author. It computes offers from a synthetic venue's inventory, exchanges the result between two independent owners, performs a separately authorized mock booking and preserves verifiable history. The authoring result connects the Football and External DRS views: the program that performs those tasks has an inspectable construction history of its own.

The engineering basis of this dossier is commit 64f2b38b0d671e4907bae8649fff8f3d2ee0751c. Functional acceptance followed an explicitly amended authoring protocol: an initial submission and three corrections remained INCOMPLETE; one separately authorized fourth correction completed the technical acceptance. The final program was then reviewed and admitted through a different owner process. A successful submission did not install itself.

Here, Root means the local authority that decides for its own system. Work is the native unit of computation; Host and Firewall bind permitted execution and effects. The author received mechanisms for connecting a domain to these interfaces, not the power to overrule them. This is a finite transfer result for a reviewed Python reference profile. It supports the architectural interpretation that construction knowledge can travel while admission authority stays with the receiving owner. It does not establish unrestricted automatic software creation or an integrated synthesis factory.

Sources: [A-S-d1result](#); [A-S-landing](#); [A-S-contract](#); [Full exact source map](#).

A02 / GATE 5 REFERENCE

HOW specifies construction; WHAT specifies the football task

The public kit separates architecture from the requested application. HOW_TO_BUILD describes native Work, local Root review, pointer-first DRS retrieval, source provenance, currentness and effect admission. Its machine interface map gives exact callable signatures, fields, file locations and hashes extracted from the supplied source. The approved snapshot is a read-only interface reference. It is not a second editable kernel or a permission to use every function found in that snapshot.

The generic numerical example computes $3 \times x + 2$, giving 23 for input 7 and 35 for input 11. A distinct two-record MOCK example demonstrates bounded effect and receipt behavior. Neither example supplies a pitch-selection algorithm. The separate WHAT_TO_BUILD brief introduces the football conditions, finite inventory, half-open intervals, integer prices, explicit consent, deterministic preference order and evidence obligations. Knowing that a natural full-size field and ninety-minute sessions are required is ordinary task disclosure; handing over a winning set of slots would be a different intervention.

The football evidence fields are explicitly labeled as a task-specific profile in CANDIDATE_OUTPUT and FOOTBALL_PROFILE. Their names are not universal Radiolaria primitives. The shared $E \rightarrow B \rightarrow C$ contract explains how an original computed result becomes published evidence and then verified consumption context. A future author can reuse that construction contract with another supported WHAT, subject to independent review. The kit therefore teaches a bounded method with concrete interfaces, rather than promising any program from an unconstrained natural-language instruction.

Input	What it contributes
HOW + machine interfaces	Construction rules, signatures, hashes and native seams
Approved source + generic examples	Read-only mechanisms; numerical and two-record MOCK demonstrations
WHAT + FOOTBALL_PROFILE	Business constraints and task-specific evidence mapping
Private examiner	Outside the author input; independent acceptance machinery

Sources: [A-S-how](#); [A-S-what](#); [A-S-interface](#); [A-S-output](#); [Full exact source map](#).

A03 / GATE 5 REFERENCE

The broker exposed seven bounded tools; execution belonged to the reviewer

The recorded author used gpt-6-astra with literal ultra effort and the default service tier. Its seven advertised broker tools could read or search approved sources, read candidate files, write or patch candidate text, inspect syntax as data and freeze a submission. A patch required the expected whole-file SHA-256 and a unique literal replacement. Submission caused the broker to calculate the package manifest. These operations provide a concrete account of what the author could see and change.

Source review and execution were separate stages. The supervisor read the initial candidate and every production change before approving a bounded runner invocation. The trusted parent created publisher and requester processes, assigned their role-specific inputs and channels, and recorded process results. The private examiner used independent source inputs and observations, rather than treating candidate-authored counters as an oracle. Later repository admission was another operation owned outside the candidate.

The white-room description is a metaphor for this declared access surface. It does not claim absence of pretraining, all platform context or all prior knowledge. The author broker is not a universal sandbox for arbitrary hostile Python. The reviewed-reference trust base includes the interpreter, approved source, runner, observer, independently frozen inputs and examiner. Same-process observation and shared worker UID are declared limitations. The earlier standalone command/exec canary discrepancy remains historical evidence; success through the narrower broker did not retroactively repair that different route.

Broker tool	Permitted operation
read_approved_source / search_approved_source	Pinned public text only
read_candidate_file	Candidate bytes and whole-file hash
write_candidate_file / patch_candidate_file	Inert candidate edits; exact patch precondition
inspect_candidate_text	Parse Python / JSON without importing or testing
submit_candidate	Freeze package; broker generates candidate.json

Sources: [A-S-tools](#); [A-S-execution](#); [A-S-authorreceipt](#); [A-S-dreport](#); [Full exact source map](#).

A04 / GATE 5 REFERENCE

The chronology preserves transport failures, incomplete trials and the accepted continuation

The history contains different kinds of events, so counting every stage as another failed author would obscure what occurred. The initial G54 Gemini transport received no response from two generation POST attempts; provider completion remained unknown and there was no submitted candidate. G54R recovered the transport and completed an initial submission plus three corrections with gemini-2.5-flash, but its final candidate failed before native Work and Root. The retained result identifies an unsupported invalidation_contexts constructor argument.

G54B investigated an Astra access route. Synthetic command/exec controls did not establish the required boundary, so no football task was delivered and no author trial started. G54B1 then used the no-environment broker route: an initial submission and three corrections reached useful two-owner computation and a mock effect, but the independent acceptance stopped at venue_offer_consumption. The separately published B1 account remains an INCOMPLETE historical result.

The C-series prepared and reviewed a corrected contract and examiner; it did not consist of new blind football authors. G54D was a new cold trial on the revised kit. Its initial submission and three corrections reached 15 of 16 native cases. G54D1 continued that same development lineage under an explicit extra correction and completed the technical checks. G55L subsequently admitted and published the engineering result. G56 documents that history with its own publication status; it does not rewrite earlier outcomes.

Stage	Recorded outcome
G54 / G54R	Gemini transport failure, then recovered initial + 3; INCOMPLETE
G54B	Boundary probe; no author task delivered
G54B1	Astra initial + 3; useful MOCK; contract conflict; INCOMPLETE
G54C-C4	Contract / examiner preparation and finite reviewed-reference qualification
G54D	New Astra cold trial, initial + 3; 15/16; INCOMPLETE
G54D1	Restored continuation, correction 4 / submission 5; accepted
G55L / G56	Engineering published / this separate publication stage

Sources: [A-S-g54initial](#); [A-S-history](#); [A-S-dreport](#); [A-S-amendment](#); [Full exact source map](#).

A05 / GATE 5 REFERENCE

The earlier contract failure tested the instructions as well as the candidate

Producer result E → Later publication B → Checked context C → Independent joins

CONTRACT MECHANISM / The old B1 examiner confused distinct representations.

G54B1 revealed an important distinction between a computation and the evidence later built around it. E is the producer Work's actual result. B publishes that result together with the identity of the completed Work and its later review. C adds the receiving side's validated status and request context. They belong to one causal history, but they cannot be identical files: the original computation cannot contain identifiers of consequences that do not yet exist.

The old examiner required the whole published body at both Work boundaries while the public instruction required that body to be assembled after producer Work and review. The revised general contract checks the causal joins instead: `payload_projection(B)` equals E; source and review references resolve to actual evidence; expected C is independently derived from the signed body, current authenticated status, original request and local policy. The consumer's own input cannot be recycled as the expected input, since that would merely compare a claim with itself.

The correction was expressed generically and exercised with a numerical profile before the next cold author. The final football profile makes its mapping explicit: `B.offer` equals E, `C.body` equals B, and the consumer receives canonical C. This is an informative limit of instruction transfer. An independent author can expose a defect in the demanded proof as well as in its own implementation. The historical B1 failure remains recorded; a versioned repair to the shared contract supports the later success. Frozen examination is valuable because changes are visible, not because an examiner is assumed infallible.

Sources: [A-S-history](#); [A-S-b1review](#); [A-S-shared](#); [A-S-output](#); [Full exact source map](#).

A06 / GATE 5 REFERENCE

G54D stopped at a genuine missing decision after the correction budget ended

The revised-kit cold trial was a real development sequence. Its initial submission failed at the Root binding used for native DRS descent. The first correction reached the offer and consumption path, then failed before effect dispatch on a capability operation/namespace mismatch. The second correction performed the main booking and history story but left a revoked publisher report and a generated source reference incompatible with the existing DRS filter. Those were concrete candidate defects, not evidence that every refusal came from a bad examiner.

The third correction fixed the issues actually reported to the author. The final matrix then exposed one remaining failure: a verified REVOKED status correctly stopped use and produced REFUSED without a booking, but the known exception branch had never invoked a local Root refusal decision. Earlier metadata and retrieval ACCEPT decisions did not provide that missing decision. The program's stop was safe in this observed case; it still failed the required evidence obligation.

The absent Root call existed before the last correction. The earlier reviewer stopped at other failures, so the specific omission was not delivered before that final permitted attempt. The process therefore could not honestly be summarized as three unsuccessful attempts to fix the same diagnosed error. The original limit was exhausted and G54D stayed INCOMPLETE. The subsequent decision was to clarify the profile's refusal obligation and explicitly allow one additional authored correction, preserving both the useful result and the failed protocol outcome.

Submission	Feedback / final finding
Initial	Invalid Root binding in native DRS descent
Correction 1	Native capability operation / namespace mismatch
Correction 2	Revoked publisher report and source-reference representation
Correction 3	Earlier items repaired; missing refusal Root first exposed

Sources: [A-S-dreport](#); [A-S-restoration](#); [A-S-clarification](#); [A-S-amendment](#); [Full exact source map](#).

A07 / GATE 5 REFERENCE

The fourth correction was a restored continuation, not a new blind trial

The amendment allowed exactly one additional correction by the author, index 4 and submission 5. The supported attempt to resume the old ephemeral thread returned no rollout found. Rather than invent continuity, the record declared RESTORED_CONTINUATION and preserved the failed resume response. The new model turn retained gpt-6-astra, ultra and default settings, but identical hidden model state was neither recovered nor claimed.

The restoration inventory names precisely what was supplied: the same public architecture/task input, the author's own final source, the feedback already delivered during G54D, and a new narrowly authorized explanation of the missing refusal. It omitted private examiner code, private case inputs, expected answers, old B1 solution code and any ready-made repair patch. The full earlier tool transcript and opaque model state were not reconstructed. The twenty-two seeded author files and their hashes identify the program being continued.

One new transport turn performed seventy-nine broker operations and submitted one corrected pack. Tool operations are reads, searches, edits, inspection and submission; they are not seventy-nine model requests. The distinction matters because the accepted claim is continuity of the recorded development lineage and exact code provenance, not a fresh test of total ignorance. The historical final reserved variation also remained the preserved G54D challenge. Its reuse after continuation supports regression and bounded variation evidence; it is not presented as a newly unseen independent benchmark.

Restored	Explicitly not restored / supplied
Exact public kit and original public task	Opaque original model state or hidden reasoning
Author's own final G54D source	Old B1 target implementation or ready-made repair
Previously delivered feedback + new diagnostic	Private examiner, fixtures or expected selections

Sources: [A-S-amendment](#); [A-S-restoration](#); [A-S-resume](#); [A-S-authorreceipt](#); [Full exact source map](#).

A08 / GATE 5 REFERENCE

The seed and recorded edits reconstruct the accepted source bytes

22-file seed → 79 broker operations → 23 authored files → 24-file package

RECORDED CONSTRUCTION / The additional file is the broker manifest.

Authorship is supported by a reproducible relationship between inputs and outputs. The reconstruction began with twenty-two exact author files from the final G54D submission. It applied the recorded broker edits in order and independently compared the resulting files with the submitted pack. This produced twenty-three author files. The broker separately generated candidate.json from the file inventory, bringing the complete package to twenty-four files. The manifest is coordinator output and is not falsely credited as model-written application logic.

The recorded changes were narrow. The production delta was in candidate_domain/run.py. The author added tests/test_status_refusal.py and updated the passport, README and source-impact explanation. Existing domain selection, policy, effect, transport and other application files retained their hashes. The source review read the complete production delta and surrounding requester flow. It found no new production imports, provider or process calls, observer access or cross-role reads; the added evidence writes used existing local helpers.

This is stronger attribution than an assistant saying that it wrote a program: exact final bytes can be derived from the known seed and observed author operations. It also keeps the support work visible. The supervisor provided task material, feedback and review machinery, but did not implement the candidate repair. The authored component tests were inspected as source and syntax; their presence does not establish a separate executed test result. Native acceptance comes from the independent runner evidence, not from file names that begin with test.

Sources: [A-S-reconstruction](#); [A-S-sourcereview](#); [A-S-patch](#); [A-S-bindings](#); [Full exact source map](#).

A09 / GATE 5 REFERENCE

The repair obtained a real refusal and bound it to the rejected source

The repaired branch preserves authenticated terminal status before refusing further use. On the known status_not_active path it revalidates retained signed evidence at the recorded observation times, then invokes a fresh local Root review whose real decision is BLOCKED_FAIL_CLOSED. The completed refusal binds the original request, actual reason, source revision and the relevant pointer/status evidence. It does not rename an earlier ACCEPT record or write a decorative rejection label.

The supplemental independent check joins those records to observed native calls in both revoked-source events. The exact event has refusal call sequence 29; the shifted event has sequence 66. These are observer call positions, not totals of Root decisions. Both native results are BLOCKED_FAIL_CLOSED, both use reason status_not_active, and their distinct request and status hashes remain linked to the actual source records. The capture records zero booking executor starts and no registry mutation for this case.

The source change was deliberately specific. It did not replace the selection algorithm, loosen the budget, manufacture consent, change the policy recipe or alter technical examiner predicates between D and D1. Unknown exceptions remain visible. A possible failure after effect start remains an uncertain outcome requiring evidence and readback; it is not rewritten as a clean no-effect refusal. That separation explains why the final correction completes an accountability path as well as a test: the local authority actually issued the decision now cited by the result.

Recorded event	Observed refusal call	Decision / reason
EXACT revoked source	29	BLOCKED_FAIL_CLOSED / status_not_active
SHIFT revoked source	66	BLOCKED_FAIL_CLOSED / status_not_active

Sources: [A-S-clarification](#); [A-S-sourcereview](#); [A-S-refusal](#); [A-S-native](#); [Full exact source map](#).

A10 / GATE 5 REFERENCE

Independent acceptance checked performance and kept evidence classes separate

After the correction, the same final candidate passed all sixteen native scenarios. The recorded matrix contains fifty-six events and seven isolated mock executor starts; one start belongs to the main three-slot booking story. Changed inputs, a second team, permutation, refusal paths and continuation exercise the implementation beyond one selected output. The examiner derives expected domain values from original runner inputs and frozen rules. A producer/consumer pair agreeing on an invented value would not satisfy that independent comparison.

The later live series is a separate experiment: three actual gemini-2.5-flash requests supplied semantic proposals for EXACT, SHIFT and VERIFY, followed by five native events and one additional mock execution. The proposals were precollected before booking. The VERIFY model did not see the subsequent receipt; the program's later verification consumed the saved history and readback. Controlled owner inputs supplied confirmation and repetition. Authoring and runtime semantic participation are therefore separate model roles.

Two supplied processes then checked the frozen D1 evidence without a new candidate run, provider call, Root decision or effect. Their outputs are byte-identical at 32,702 bytes. Altered-evidence controls are pure checking of data, not additional native attacks. The scope is bounded: reviewed-reference trust, synthetic inventory and MOCK action, no arbitrary hostile Python guarantee or physical venue integration. Acceptance is substantial because each class of evidence answers its own question; adding unlike counters together would make the result less informative.

Evidence class	Recorded scope
D1 native	16 cases / 56 events / 7 isolated mock executor starts
D1 live	3 provider requests / 5 events / 1 further mock executor start
D1 supplied	2 independent processes / 32,702-byte identical output

Sources: [A-S-native](#); [A-S-d1result](#); [A-S-live](#); [A-S-supplied](#); [Full exact source map](#).

A11 / GATE 5 REFERENCE

Engineering admission published the reviewed bytes through a separate owner process

A technically accepted candidate remained an external result until its source was admitted. G55 assembled an exact proposal, portable evidence and the finite acceptance matrix. External Work review checked those bytes before G55L used the unchanged repository R1 process. The actual engineering commit is 64f2b38b0d671e4907bae8649fff8f3d2ee0751c, with parent 5cef2c52fd805daedec958109caf9d1806899eb4 and tree cacb14a9574c350fb33239680abe0c3a5ede7b5e. The ordinary push and independent remote readback are recorded as completed.

The admitted change adds 212 paths while preserving all 4,367 earlier tracked files and their modes. Within those additions are thirteen shared source/test/checkpoint files and the exact twenty-four-file authored pack, plus the kit and evidence. Core, Root, Host and Firewall source were not rewritten to make the target story pass. Placing the pack in the repository does not make its manifest a runtime permission or automatically execute it.

The installed offline proof returned the expected 37,389-byte canonical result. This is a repository-level verification report, distinct from the earlier 32,702-byte D1 supplied output; their different sizes describe different report generations. Historical kit and preparation documents still contain their original PENDING labels. The current status bridge uses later landing receipts instead of editing those frozen files. Engineering acceptance and publication are complete; the separate G56 documentation publication and final closure rows must receive their own actual completion evidence.

Admission fact	Value
New paths	212
Earlier paths / modes preserved	4,367
Installed proof	37,389 bytes; expected canonical SHA-256
Engineering	Accepted, committed and independently read back
G56 documentation / full closure	Separate publication acceptance still required

Sources: [A-S-landing](#); [A-S-preserved](#); [A-S-contract](#); [A-S-bindings](#); [Full exact source map](#).

A12 / GATE 5 REFERENCE

The published kit is a starting contract for another reviewed extension

A developer should begin with the current kit entry, then read HOW_TO_BUILD, SHARED_EVIDENCE_CONTRACT and the machine interface map. SOURCE_INVENTORY identifies the supplied code and schema resource; CANDIDATE_OUTPUT describes the bounded return package. The separate WHAT_TO_BUILD and FOOTBALL_PROFILE are the demonstrated application. The accepted football pack is useful as a disclosed example, but it must not be silently included in a future blind-author input. The auditor's historical corpus and private examiner are different deliverables.

A concrete extension point is the exact run_peer_v01 interface: role, input_data, state_root, output_root, channel and bootstrap are keyword arguments supplied by the approved runner. A reviewed local body profile supplies closed body/scope fields and local validate_body, validate_pointer and validate_binding functions. Neither a remote payload nor a signature selects executable code. A new implementation needs its own source review, supported runner and separate admission; this publication conveys construction knowledge, not the original owner's credentials or installation authority.

The observed transfer result is one new working domain created through this bounded contract. The architectural interpretation is that an extension's meaning and implementation can change while the receiving system retains the decision boundaries that govern use. Reflective Improvement Radiolaria, Cross-Domain Transfer Radiolaria and an integrated Capability Synthesis Factory are future directions, explicitly deferred beyond this reference. They may build on the same separation of proposals and decisions, but they are not implemented merely because a model successfully authored this package.

Machine anchor	Exact supported surface
Peer endpoint	run_peer_v01(*, role, input_data, state_root, output_root, channel, bootstrap)
Reviewed body profile	Closed fields; validate_body / validate_pointer / validate_binding
Package manifest	g53.candidate.v01; installable_now=false; exact byte hashes
Future extensions	Reflective Improvement / Cross-Domain Transfer / Factory: deferred

Sources: [A-S-kitstart](#); [A-S-how](#); [A-S-interface](#); [A-S-inventory](#); [Full exact source map](#).